

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_random_replay.
- Best TSPLIB holdout gap: tsplib_random_replay.
- Best synthetic holdout gap: tsplib_random_replay.

Run Metadata

- run_name: run_20260513_132232_e
- started_at_local: 2026-05-13 13:22:32
- finished_at_local: 2026-05-13 13:32:39
- duration_hhmm: 00:10
- duration_seconds: 606.772
- seed_offset: 4000
- replicate_label: e
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.876497	0.56	-0.028414
tsplib_random_replay	random	score_only	False	0.174852	0.0	0.111269	0.783231	0.76	0.010286
tsplib_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.885112	0.56	-0.028138
tsplib_failure_replay_compression	failure	novelty_gate	True	0.213387	0.064916	0.159398	0.0	0.56	0.0

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.876497, and final complexity 0.56.
- Adaptation efficiency -0.028414 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.174852, synthetic 0.0, combined 0.111269.
- Accepted-epoch count 2, mean accepted code novelty 0.783231, and final complexity 0.76.
- Adaptation efficiency 0.010286 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.174852 across 7 instances; family means: ch=0.119618, kroD=0.117169, pcb=0.275848, pr=0.440555, rd=0.065234, st=0.085926.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3060, "family": "a", "name": "a280", "optimality_gap": 0.186506}, {"best_known_cost": 7542, "cost": 8060, "family": "berlin", "name": "berlin52", "optimality_gap": 0.068682}, {"best_known_cost": 14379, "cost": 15319, "family": "lin", "name": "lin105", "optimality_gap": 0.065373}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.885112, and final complexity 0.56.
- Adaptation efficiency -0.028138 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.

- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

Judge Appendix

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Summary of Replay Conditions in TSP Benchmark Suite

Condition	Replay Mode	Compression	Mean Held-out TSPLIB Gap	Mean Synthetic Holdout Gap	Mean Transfer Gap	Final Complexity	Code Novelty (mean)	Notes on Metrics and Transfer
tsplib_no_replay	None	No	0.213	0.065	0.159	0.56	0.876	Baseline gaps relatively high.
tsplib_random_replay	Random	No	**0.175**	**0.0**	**0.111**	0.76	0.783	Best transfer and generalization performance; lower code novelty than no replay.
tsplib_failure_replay	Failure	No	0.213	0.065	0.159	0.56	0.885	Same gaps as no replay; high code novelty but no transfer improvement.
tsplib_failure_replay_compression	Failure + Compression	Yes	0.213	0.065	0.159	0.56	0.0	No code novelty; no gap improvement over failure replay without compression.

Detailed Interpretation

- ****Transfer metrics (held-out TSPLIB and synthetic holdout gaps):****
- ****tsplib_random_replay**** achieves the lowest held-out TSPLIB gap (0.175) and synthetic holdout gap (0.0), representing the best transfer/generalization.
- Other replay conditions, including no replay and failure replay (with or without compression), show the same higher held-out and synthetic gaps (0.213 and 0.065 respectively).
- ****Code novelty vs. Transfer:****
- ****tsplib_random_replay**** shows a notable reduction in mean code novelty (~0.783) compared to no replay (~0.876) and failure replay (~0.885), while achieving better transfer performance.
- This indicates that lower lexical/code novelty does not preclude improved transfer/generalization.
- Failure replay (both with and without compression) maintains high code novelty but does not improve transfer gaps relative to no replay.

- Compression-aware failure replay notably results in zero code novelty but offers no improvement in transfer, suggesting compression reduces novelty without benefiting algorithmic performance.
- **Replay modes:**
- **No replay:** Baseline with no replay experience; associated with largest transfer gaps.
- **Random replay:** Sampling replayed experiences randomly; yields best transfer performance and lowers code novelty.
- **Failure replay:** Replaying failure cases only; no transfer improvement compared to no replay.
- **Failure replay with compression:** Adds compression pressure to failure replay; further lowers code novelty to zero but transfer gaps remain unimproved.
- **Complexity and behavior:**
- Random replay leads to higher final complexity (0.76) and clustered constructor behavior profile, correlated with better transfer.
- Other conditions have moderate complexity (0.56) and balanced behavior.
- **Adaptation efficiency:**
- Random replay uniquely shows positive adaptation efficiency (0.0103), indicating better learning dynamics.

Conclusions

- **Best transfer and generalization are achieved under `tsplib_random_replay`, confirming the efficacy of random replay over no replay or failure-only replay in this benchmark.**
- **Improved transfer under random replay occurs despite a decline in code novelty, indicating code novelty is not a driver of transfer performance here.**
- **Failure replay (with or without compression) maintains high code novelty but does not improve transfer metrics versus no replay, challenging the assumption that lexical novelty correlates with algorithmic innovation.**
- **Compression-aware replay suppresses code novelty entirely without providing transfer benefits, suggesting it reduces exploration without enhancing solution quality.**

Hence, random replay is the most effective replay strategy for this TSP benchmark suite, yielding better optimality gaps and transfer with moderately lower code novelty.